222720: metagabbro, Oberwyl Hill

(Forest Belle Gabbro, Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Menzies (SH 51-5), Ballard (3039)
MGA Zone 51, 257846E 6784240N

Sampled on 7 April 2018

This sample was collected from low, rubbly outcrop (Fig. 1) on the northeastern flank of Oberwyl Hill, about 3.8 km southwest of Charlie Well, 1.7 km east-northeast of Peppertree Well, and 0.5 km southeast of the Leonora – Mount Ida Road.

Geological context

The unit sampled is the Forest Belle Gabbro, within the eastern limb of the Kurrajong Anticline in the Mount Ida greenstone belt of the Kalgoorlie Terrane (Rattenbury, 1995; Wyche, 2003). The north-trending metagabbro is about 10 km long and 1 km wide, and consists of coarse-grained metagabbro with an ophitic texture in which cm-scale clinopyroxene oikocrysts, typically altered to green amphibole, enclose tabular plagioclase crystals up to 5 cm long (Wyche, 2003). The sample was collected for geochronology near the eastern contact of the metagabbro with Archean amphibolite and basaltic rocks. The metagabbro is strongly foliated and cut by auriferous quartz veins near its western contact with the undated Copperfield Monzogranite (Wyche, 2003). About 4.4 km to the north, quartzite interleaved with mafic–ultramafic rocks of the Walter Williams Formation of the Kalgoorlie Group (Swager et al., 1990; Hill et al., 2001), yielded a maximum depositional age of 2764 ± 8 Ma (GSWA 222712, Wingate et al., 2022).

Petrographic description

The sample is a medium- to coarse-grained, strongly altered metagabbro (Fig. 1), consisting of about 60–65% altered plagioclase, 30–35% actinolite, 3–4% epidote, and minor biotite, apatite, epidote, magnetite and titanite. Poorly defined, optically continuous patches of albite grains and minor fine-grained quartz are interpreted to be ex-plagioclase grains, which range in size up to 5 mm. Rare multiple twinning and a crudely undulose blotchy texture within these patches is suggestive of hydrothermal albite. Green actinolite forms mainly elongate, irregular aggregates up to 12 mm long that are crudely aligned and anastomose around the margins of altered plagioclase patches. Within the aggregates, actinolite varies in grain size, from microcrystalline to 1.5 mm long, and habit, from strongly aligned subhedral laths, to decussate aggregates, to radiating acicular needles. Biotite laths up to 0.3 mm are distributed throughout the sample, and one aggregate of decussate subhedral laths up to 0.3 mm long was observed in the thin section. Anhedral to subhedral epidote up to 0.1 mm in size overprints some albitized plagioclase. Fine-grained subhedral magnetite contains interstitial carbonate minerals and smaller grains of hematite-altered magnetite. Titanite forms irregular aggregates and anhedral grains up to 0.5 mm in size, some including minor magnetite. The sample has a weak foliation defined by actinolite but lacks a penetrative tectonic fabric. The sample is interpreted as a hydrothermally altered metagabbro, which originally consisted mainly of plagioclase and pyroxene.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown, partially iron-stained, and mostly subhedral. The crystals are up to 300 µm long, and equant to elongate, with aspect ratios up to 6:1. In cathodoluminescence (CL) images, broad planar zoning is ubiquitous, and some crystals exhibit patchy zoning or alteration along fractures. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 25–26 June 2020, using SHRIMP-B. Twelve analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.70% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.27% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.39%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-two analyses were obtained from 21 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Figure 1. **Outcrop image for sample 222720: metagabbro, Oberwyl Hill**

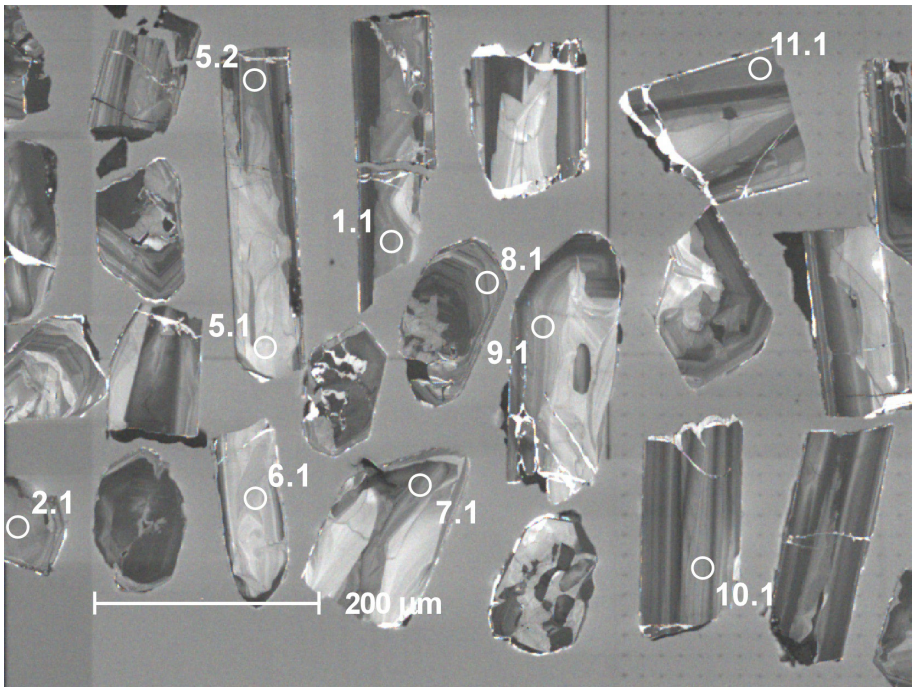

Figure 2. **Cathodoluminescence image of representative zircons from sample 222720: metagabbro, Oberwyl Hill. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 222720: metagabbro, Oberwyl Hill

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
I	2	2.1	34	348	10.09	0.058	1.926	0.042	0.18702	0.00215	1.927	0.042	0.18650	0.00222	2695	48	2712	20	0.6
I	17	16.1	81	215	2.66	0.391	1.916	0.031	0.19038	0.00155	1.923	0.031	0.18691	0.00182	2699	36	2715	16	0.6
I	3	3.1	60	154	2.59	0.338	1.974	0.035	0.18996	0.00177	1.981	0.035	0.18695	0.00204	2635	39	2716	18	3.0
I	5	5.1	6	122	18.97	0.346	1.940	0.099	0.19013	0.00520	1.947	0.099	0.18706	0.00606	2673	112	2717	53	1.6
I	1	1.1	53	101	1.90	0.000	1.816	0.033	0.18757	0.00174	1.816	0.033	0.18757	0.00174	2827	42	2721	15	-3.9
I	20	19.1	15	112	7.69	-0.307	1.876	0.065	0.18517	0.00341	1.871	0.065	0.18790	0.00391	2761	78	2724	34	-1.4
I	14	13.1	63	49	0.78	-0.173	1.934	0.034	0.18640	0.00171	1.931	0.034	0.18793	0.00184	2690	39	2724	16	1.2
I	4	4.1	63	95	1.51	0.036	1.895	0.034	0.18831	0.00174	1.895	0.034	0.18799	0.00177	2731	40	2725	16	-0.2
I	16	15.1	59	107	1.81	0.000	1.934	0.035	0.18807	0.00178	1.934	0.035	0.18807	0.00178	2687	40	2725	16	1.4
I	21	20.1	18	488	27.86	0.670	2.025	0.063	0.19410	0.00328	2.038	0.063	0.18814	0.00426	2573	66	2726	37	5.6
I	8	7.1	73	88	1.21	-0.066	1.857	0.032	0.18757	0.00166	1.855	0.032	0.18816	0.00171	2779	39	2726	15	-1.9
I	12	11.1	89	182	2.05	0.097	1.901	0.029	0.18927	0.00220	1.903	0.030	0.18841	0.00224	2722	35	2728	20	0.2
I	13	12.1	122	37	0.31	0.000	1.953	0.041	0.18916	0.00133	1.953	0.041	0.18916	0.00133	2666	46	2735	12	2.5
I	11	10.1	78	60	0.77	0.189	2.042	0.033	0.19106	0.00154	2.046	0.033	0.18937	0.00167	2565	34	2737	15	6.3
I	22	21.1	50	129	2.60	-0.412	1.861	0.037	0.18755	0.00193	1.853	0.037	0.19122	0.00228	2782	45	2753	20	-1.1
I	10	9.1	67	96	1.45	-0.135	1.949	0.034	0.19041	0.00170	1.946	0.034	0.19161	0.00180	2673	38	2756	15	3.0
I	9	8.1	30	442	14.74	-0.294	1.943	0.047	0.19086	0.00245	1.937	0.047	0.19347	0.00277	2683	53	2772	23	3.2
P	7	6.1	12	119	10.00	0.886	1.980	0.071	0.18171	0.00357	1.997	0.072	0.17384	0.00507	2617	78	2595	49	-0.8
D	19	18.1	13	435	33.69	1.765	1.954	0.071	0.19840	0.00400	1.989	0.074	0.18272	0.00672	2626	80	2678	61	1.9
D	6	5.2	251	683	2.72	0.465	6.224	0.194	0.18943	0.00137	6.253	0.195	0.18529	0.00163	956	28	2701	14	64.6
D	15	14.1	8	220	26.22	-1.395	2.508	0.112	0.18592	0.00521	2.474	0.112	0.19830	0.00800	2189	84	2812	66	22.2
D	18	17.1	10	247	24.77	1.311	2.333	0.095	0.21341	0.00799	2.364	0.098	0.20179	0.00968	2274	79	2841	78	19.9

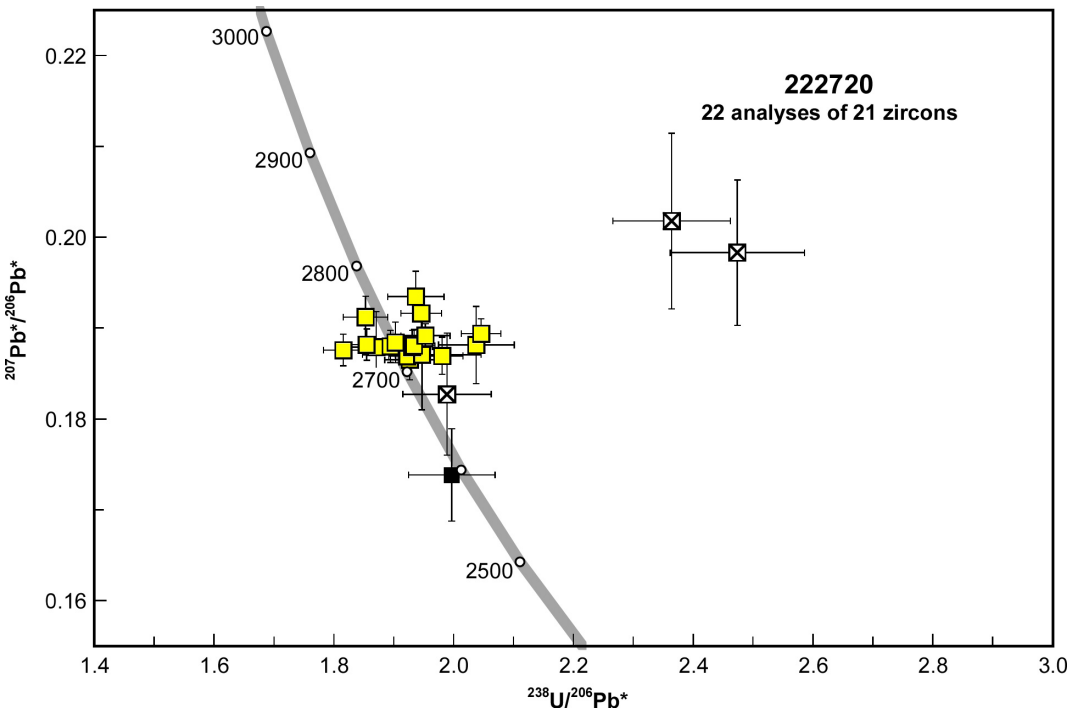

Figure 3. U–Pb analytical data for zircons from sample 222720: metagabbro, Oberwyl Hill. Yellow squares indicate Group I (magmatic zircons); black square indicates Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >7% or high common Pb). One highly discordant analysis in Group D is not shown

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Four analyses are >7% discordant and/or indicate high common Pb (*f*204 >1%). The dates obtained from these four analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 18 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 17 analyses of 17 zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2731 ± 8 Ma (MSWD = 0.69).

Group P comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2595 ± 49 Ma (1σ).

The date of 2731 ± 8 Ma for the 17 analyses in Group I is interpreted as the igneous crystallization age of the gabbro. Highly variable Th concentrations and Th/U ratios are consistent with the zircons having formed during igneous crystallization of the gabbro (e.g. Lanyon et al., 1993). The date of 2595 ± 49 Ma (1σ) for the single analysis in Group P is interpreted to reflect loss of radiogenic Pb.

References

Hill, RET, Barnes, SJ and Dowling, SE 2001, Komatiites of the Norseman–Wiluna Greenstone Belt, Western Australia — a field guide: Geological Survey of Western Australia, Record 2001/10, 71p.

Lanyon, R, Black, LP and Seitz, H 1993, U-Pb zircon dating of mafic dykes and its application to the Proterozoic geological history of the Vestfold Hills, East Antarctica: Contributions to Mineralogy and Petrology, v. 115, p. 184–203.

Rattenbury, MS 1995, Ballard, WA Sheet 3039 (2nd edition): Geological Survey of Western Australia, 1:100 000 Geological Series.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Swager, CP, Griffin, TJ, Witt, WK, Wyche, S, Ahmat, AL, Hunter, WM and McGoldrick, PJ 1990, Geology of the Archaean Kalgoorlie Terrane — an explanatory note: Geological Survey of Western Australia, Record 1990/12, 26p.

Wingate, MTD, Lu, Y, Fielding, IOH and De Paoli, M 2022, 222712: quartzite, Lawlers – Mount Ida Road; Geochronology Record 1855: Geological Survey of Western Australia, 4p.

Wyche, S 2003, Menzies, WA (2nd edition): Geological Survey of Western Australia, 1:250 000 Geological Series Explanatory Notes, 38p.

Recommended reference for this publication

Wingate, MTD, Lu, Y, Fielding, IOH and De Paoli, M 2022, 222720: metagabbro, Oberwyl Hill; Geochronology Record 1856: Geological Survey of Western Australia, 4p.

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